

Appendix A Construction of Nutrient Complementarity Rules

NCKG adds a directed **complements** edge from ingredient i to j when a physiological rule states that a nutrient abundant in i (tagged **high_A**) enhances the absorption or function of a nutrient abundant in j (tagged **high_B**). Table A1 lists the seven rules used, all within the dataset’s six-dimensional nutritional space.

Although the rule set is small, it produces the 91,856 **complements** edges in the KG. Each rule operates by connecting every ingredient with the **high_A** tag to every ingredient with the **high_B** tag. This pairwise mapping generates a total number of edges equal to the sum of $|\mathcal{V}_{\text{high_A}}| \times |\mathcal{V}_{\text{high_B}}|$ across all seven rules.

Table A1. The basic nutrient complementarity rules used in NCKG.

Nutrient Pair	Biological Mechanism	Example Pairing
Sodium $\rightarrow$ Carbs	Intestinal glucose absorption is mediated by Na ⁺ -coupled transporters (SGLT1) [1].	Soy Sauce $\rightarrow$ Rice
Sodium $\rightarrow$ Protein	Amino acid uptake in the gut relies on Na ⁺ -dependent cotransport systems [2].	Salt $\rightarrow$ Chicken
Carbs $\rightarrow$ Protein	Insulin secretion triggered by carbohydrates promotes amino acid uptake and protein synthesis [3].	Potato $\rightarrow$ Beef
Protein $\rightarrow$ Carbs	Co-ingestion of protein with carbohydrates accelerates postprandial glycogen replenishment [4].	Egg $\rightarrow$ Toast
Energy $\rightarrow$ Protein	Adequate non-protein energy supply prevents the oxidation of amino acids for fuel [5].	Butter $\rightarrow$ Steak
Fiber $\rightarrow$ Carbs	Viscous dietary fiber delays gastric emptying, attenuating postprandial glycemic excursions [6].	Broccoli $\rightarrow$ Pasta
Fat $\rightarrow$ Carbs	Dietary fat slows carbohydrate digestion and absorption, smoothing the glycemic response curve [7].	Olive Oil $\rightarrow$ Bread

Appendix B Prompt Template Repository

To mitigate the potential bias introduced by the model’s sensitivity to specific natural language phrasings, we employed a diverse set of prompt templates. In our evaluation, user profiles from NCKG are parameterized using the following variables:

- **<Health_Status>**: The user’s physiological condition (e.g., Diabetes, Hypertension).
- **<Nutrient_Targets>**: The exact quantitative dietary constraints (e.g., Energy 450 kcal, Sodium < 400 mg).

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- <Likes>: Preferred ingredients to include.
- <Dislikes>: Ingredients to strictly avoid.

Appendix B.1 Task A: Discriminative Retrieval

The following 7 prompt templates were utilized to instruct the LLM to retrieve and rank existing recipes:

1. “Can you suggest something to eat? To manage my <Health_Status>, I need a meal under <Nutrient_Targets>. I prefer <Likes> and must avoid <Dislikes>.”
2. “As a patient managing <Health_Status>, I require a meal suggestion satisfying <Nutrient_Targets>. Please retrieve options with <Likes> and exclude <Dislikes>.”
3. “I need a custom recipe for my <Health_Status>. It must strictly hit <Nutrient_Targets>. I’d love it to feature <Likes>, but definitely no <Dislikes>.”
4. “My diet restricts me to <Nutrient_Targets> due to <Health_Status>. Find me a <Likes> dish without any <Dislikes>.”
5. “Search existing recipes. User Profile: <Health_Status>. Constraints: <Nutrient_Targets>. Ingredients to include: <Likes>. Ingredients to avoid: <Dislikes>.”
6. “Act as my personal nutritionist. Based on my <Health_Status> condition, rank the best recipes that provide <Nutrient_Targets>, featuring <Likes> but skipping <Dislikes>.”
7. “I’m looking for a dish that fits my <Health_Status> dietary goals. Do you have anything matching <Nutrient_Targets>? I enjoy <Likes> but exclude <Dislikes> please.”

Appendix B.2 Task B: Constrained Generation

The following 7 prompt templates were utilized to instruct the LLM to generate novel recipes:

1. “I need a custom recipe for my <Health_Status>. It must strictly hit <Nutrient_Targets>. I’d love it to feature <Likes>, but definitely no <Dislikes>.”
2. “Design a new recipe tailored for <Health_Status> management. The output must perfectly match <Nutrient_Targets>. Use <Likes> and strictly avoid <Dislikes>.”
3. “Create a creative dish for a <Health_Status> diet. Ingredients needed: <Likes>. Prohibited: <Dislikes>. Exact nutrition targets: <Nutrient_Targets>.”
4. “Please develop a full recipe from scratch for my <Health_Status>. It should meet <Nutrient_Targets>. I enjoy <Likes>, but exclude <Dislikes> entirely.”
5. “Generate a recipe where computed nutrition is exactly <Nutrient_Targets>. Context: <Health_Status>. Primary ingredient: <Likes>. Forbidden: <Dislikes>.”
6. “I’m strictly tracking macros for my <Health_Status>. Create a new meal plan hitting <Nutrient_Targets>, using <Likes> but absolutely no <Dislikes>.”
7. “If I want to manage <Health_Status> with <Likes> but no <Dislikes>, what new recipe can you generate that provides exactly <Nutrient_Targets>?”

Appendix C Extended Experimental Analysis and Details

Appendix C.1 TPGO Convergence Analysis

Table A2 reports the convergence rate of TPGO (Algorithm 1) across all Task B instances, stratified by the most restrictive nutrient dimension (defined as the nutrient with the largest initial relative deviation from the target).

Table A2. TPGO convergence rate by most restrictive nutrient.

Nutrient dimension	Instance proportion	Convergence rate	Discard rate
Energy (kcal)	4.0%	92.1%	7.9%
Protein (g)	4.0%	91.4%	8.6%
Fat (g)	3.0%	94.2%	5.8%
Carbohydrates (g)	85.0%	95.0%	5.0%
Fiber (g)	2.0%	93.7%	6.3%
Sodium (mg)	2.0%	90.5%	9.5%
Overall	100.0%	94.6%	5.4%

Appendix C.2 Sensitivity Analysis and Experimental Details

Random Seed for Task A Sampling. We repeated the candidate sampling for Task A with three different random seeds (42, 123, 999). The standard deviation of Recall@3 across these seeds is approximately 0.008, confirming the robustness mentioned in Section 4.2.

Task B Trigger Threshold γ . Varying γ from 0.3 to 0.7 changes the Task B proportion from 12% to 58%; at our default $\gamma = 0.5$, Task B accounts for 37% of the dataset, with a standard deviation of $< 1.5\%$ across random seeds.

Retrieval Weight w . Varying w from 0.6 to 0.8 yields a Kendall’s $\tau > 0.92$ and changes fewer than 5% of top-3 rankings, confirming the robustness of the retrieval scoring function.

TPGO Initialization Strategies. We tested three initialization strategies for the quantity vector $\mathbf{q}^{(0)}$: (i) uniform 100g, (ii) uniform 50g, and (iii) category-specific defaults (100g per ingredient, except 5g for spices). The convergence rate varied by less than 2% (ranging from 93.1% to 95.2%), confirming stability.

Stability of TPGO Convergence. Running TPGO with 10 different random initializations on 500 queries yields a coefficient of variation below 0.05 for 94% of the queries, consistent with the results reported in Section 5.3.

Context Window Management for Task A. To evaluate retrieval performance in Task A, all up to $K = 100$ candidate recipes sampled via SPARQL (comprising recipe names, ingredient lists, and exact six-dimensional nutritional profiles) were formatted as a JSON array and appended to the model’s prompt. This structured candidate list introduces approximately 8k to 12k tokens per query, which comfortably fits within the extended context windows (e.g., 128k) of the evaluated modern LLMs.

Discussion on Baseline Discrepancies. While Khamesian et al. [8] reported extremely low error rates (e.g., 1.55%) for NutriGen, their evaluation optimized and measured solely against a single target (total caloric intake). In contrast, Nutr20K enforces strict and simultaneous 6-dimensional nutritional constraints (energy, protein, fat, carbohydrates, fiber, sodium). The substantial performance degradation of NutriGen on our dataset (NPR 28.6%) does not contradict their findings but rather highlights the exponential difficulty of multi-target optimization: prompt-engineering approaches that succeed in 1D constraints often struggle to balance 6 coupled nutritional targets simultaneously without structured algorithmic intervention.

Appendix D Model Access

Table A3 lists the Hugging Face identifiers of all open-weight models evaluated on Nutr20K. All models are publicly available under their respective licenses.

Table A3. Evaluated models and their Hugging Face identifiers.

Model	Identifier	Params
Llama-3.1-8B-Instruct	meta-llama/Llama-3.1-8B-Instruct	8B
Qwen2.5-7B-Instruct	Qwen/Qwen2.5-7B-Instruct	7B
GLM-4-9B-Chat	THUDM/glm-4-9b-chat	9B
Mixtral-8x22B-Instruct	mistralai/Mixtral-8x22B-Instruct-v0.1	141B
Llama-3.3-70B-Instruct	meta-llama/Llama-3.3-70B-Instruct	70B
Qwen2.5-72B-Instruct	Qwen/Qwen2.5-72B-Instruct	72B
DeepSeek-V3	deepseek-ai/DeepSeek-V3	671B

References

1. Wright, E.M., Loo, D.D.F., Hirayama, B.A.: Biology of human sodium glucose transporters. *Physiol. Rev.* **91**(2), 733–794 (2011)
2. Bröer, S.: Amino acid transport across mammalian intestinal and renal epithelia. *Physiol. Rev.* **88**(1), 249–286 (2008)
3. Margolis, L.M., Allen, J.T., Hatch-McChesney, A., Pasiakos, S.M.: Coingestion of carbohydrate and protein on muscle glycogen synthesis after exercise: a meta-analysis. *Med. Sci. Sports Exerc.* **53**(2), 384–393 (2021)
4. Zawadzki, K.M., Yaspekis, B.B., Ivy, J.L.: Carbohydrate-protein complex increases the rate of muscle glycogen storage after exercise. *J. Appl. Physiol.* **72**(5), 1854–1859 (1992)
5. Pellett, P.L.: Protein requirements in humans. *Am. J. Clin. Nutr.* **51**(5), 723–737 (1990)
6. Lattimer, J.M., Haub, M.D.: Effects of dietary fiber and its components on metabolic health. *Nutrients* **2**(12), 1266–1289 (2010)
7. Collier, G., O’Dea, K.: The effect of coingestion of fat on the glucose, insulin, and gastric inhibitory polypeptide responses to carbohydrate and protein. *Am. J. Clin. Nutr.* **37**(6), 941–944 (1983)
8. Khamesian, S., Arefeen, A., Carpenter, S.M., Ghasemzadeh, H.: NutriGen: Personalized meal plan generator leveraging large language models to enhance dietary and nutritional adherence. In *Proc. 47th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, pp. 1–7, Copenhagen, Denmark, 2025.